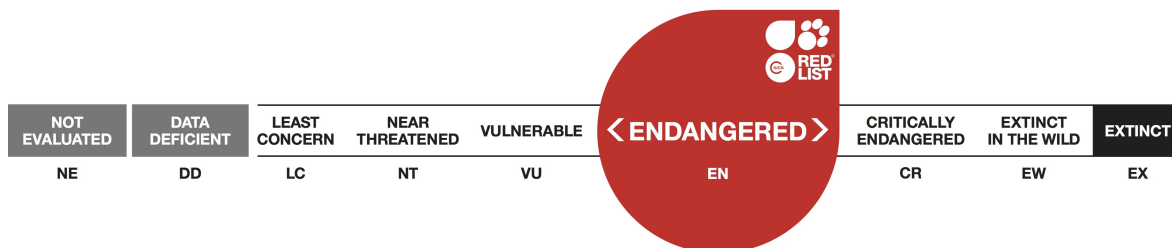


## *Nestor notabilis*, Kea

Assessment by: BirdLife International



View on [www.iucnredlist.org](http://www.iucnredlist.org)

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## Taxonomy

| Kingdom  | Phylum   | Class | Order          | Family      |
|----------|----------|-------|----------------|-------------|
| Animalia | Chordata | Aves  | Psittaciformes | Strigopidae |

**Taxon Name:** *Nestor notabilis* Gould, 1856

**Common Name(s):**

- English: Kea

**Taxonomic Source(s):**

Turbott, E.G. 1990. *Checklist of the Birds of New Zealand*. Ornithological Society of New Zealand, Wellington.

**Identification Information:**

48 cm. Inquisitive alpine parrot. Olive-green with scarlet underwings and rump. Dark-edged feathers. Dark brown bill, cere, iris, legs and feet. Male has longer bill. Juvenile has yellow cere, eye-ring and on bill. **Similar spp.** Kaka *N. meridionalis* is a lowland species, smaller, darker with crimson underparts. **Voice** Loud keee-aa.

## Assessment Information

**Red List Category & Criteria:** Endangered A2be+4be [ver 3.1](#)

**Year Published:** 2017

**Date Assessed:** October 1, 2017

**Justification:**

This species is believed to be declining rapidly because of predation by introduced mammals and a variety of anthropogenic threats. For this reason it is listed as Endangered.

**Previously Published Red List Assessments**

2016 – Vulnerable (VU)

<http://dx.doi.org/10.2305/IUCN.UK.2016-3.RLTS.T22684831A93048746.en>

2012 – Vulnerable (VU)

<http://dx.doi.org/10.2305/IUCN.UK.2012-1.RLTS.T22684831A38993727.en>

2008 – Vulnerable (VU)

2006 – Vulnerable (VU)

2004 – Vulnerable (VU)

2000 – Vulnerable (VU)

1994 – Lower Risk/near threatened (LR/nt)

1988 – Lower Risk/least concern (LR/lc)

## Geographic Range

### Range Description:

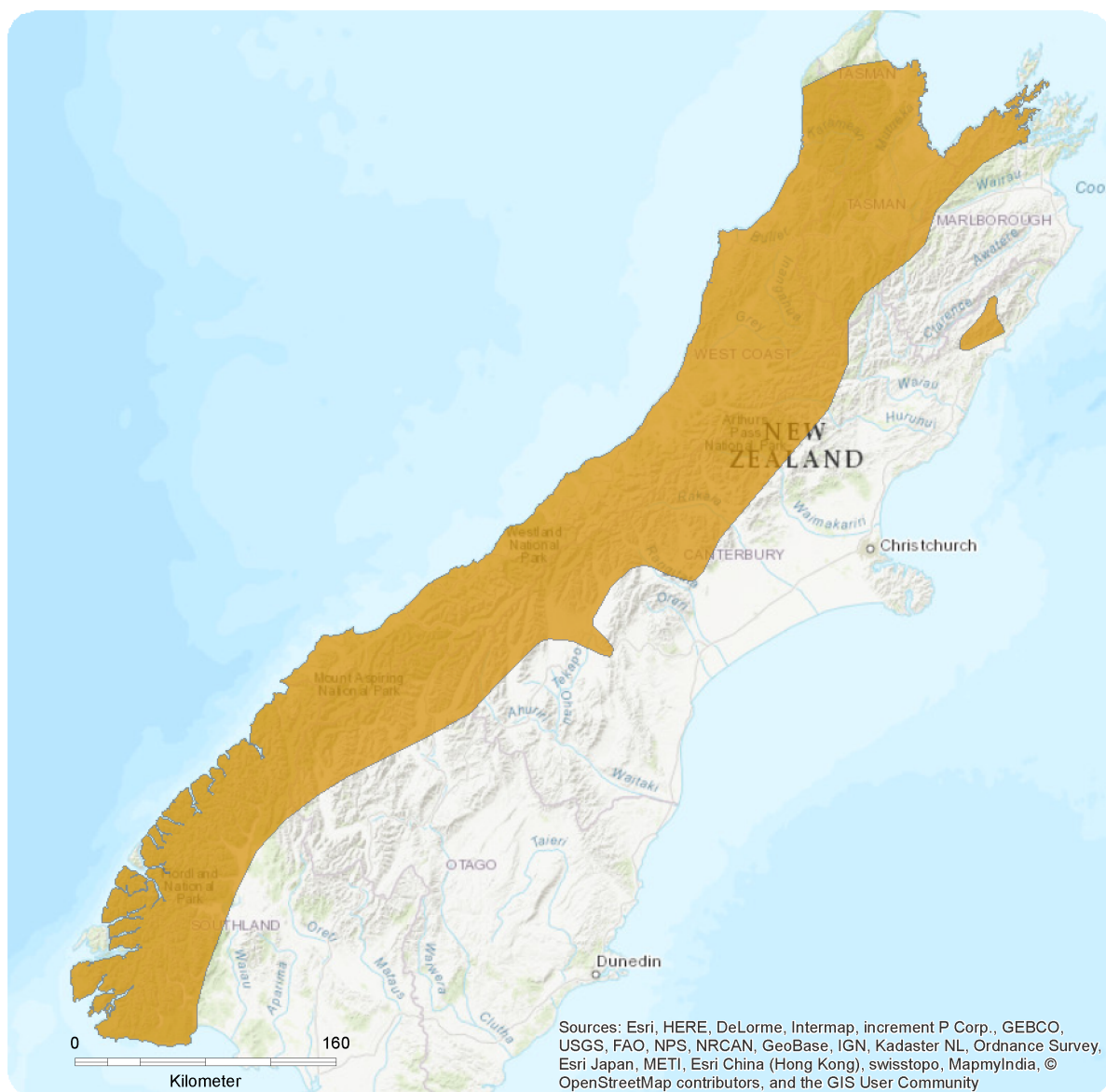
*Nestor notabilis* occurs on South Island, **New Zealand**. The population is sparsely distributed across a range of approximately 3.5 million hectares from Kahurangi to Fiordland, and including the Kaikoura Ranges. The population is a fraction of what it once was, largely due to persecution between the late 1860s and early 1970s, although pockets of high population densities persist in some areas, such as around Arthur's Pass and South Westland (Orr-Walker *et al.* 2015). The current population has been estimated to number 6,000 individuals (Kemp 2013) and continues to decline rapidly.

### Country Occurrence:

**Native:** New Zealand

# Distribution Map

*Nestor notabilis*



## Range

Extant (resident)

## Compiled by:

BirdLife International and Handbook of the Birds of the World (2016)



The boundaries and names shown and the designations used on this map do not imply any official endorsement, acceptance or opinion by IUCN.



## Population

A conservative estimate of one adult female per 2,000 hectares of forest gives a total population of 4,000 mature individuals in the population. Productivity estimates predict one juvenile for every breeding pair, giving a total population of about 6,000 birds (Kemp 2013).

### Trend Justification

Recent survey data indicate that Kea have undergone substantial recent population declines. For example, density in the upland beech forest of Nelson Lakes National Park in 2011 was approximately one adult female Kea per 2,750 hectares, down from about one per 550 ha in 1998 (Adams *et al.*, 2011; Kemp 2013): this represents an 80% decline in density over 13 years or just over one generation. There are also numerous anecdotal reports of decreases from other unmanaged areas (Orr-Walker *et al.* 2015). Although densities in other populations are much higher than at Nelson Lakes, it is likely that Kea in areas not subject to predator control are continuing to decline. The total population decline over the last three generations (36 years) is likely to have been more than 50% but less than 80%.

**Current Population Trend:** Decreasing

## Habitat and Ecology (see Appendix for additional information)

It mostly inhabits high-altitude forest and alpine basins, although birds will often frequent and nest in coastal lowland flats. Its foraging habitat includes all types of native forest, sub-alpine scrub, tussock and herb-field. It mostly feeds on berries and shoots, although many have adapted to feeding at refuse dumps and ski-fields. It nests in holes, under logs or in rocky crevasses, mainly within forest. It usually incubates up to four eggs. Males feed the females during incubation and after hatching. Birds breed after three or more years. The oldest recorded wild bird was at least 22 years of age (T. Orr-Walker *in litt.* 2016).

**Systems:** Terrestrial

## Threats (see Appendix for additional information)

Up until its partial protection in the early 1970s, over 150,000 were shot in a bounty scheme, established because rogue individuals were found to be attacking sheep as a source of fat. Deforestation for pasture has placed pressure on the species, and farmers still kill an unknown number of birds each year (Mosen 2009). Introduced mammalian predators such as stoats *Mustela erminea*, cats and brush-tailed possums *Trichosurus vulpecula* have spread into most of the species's range. Episodic, high mortality events are thought to be associated with “plagues” of stoats *Mustela erminea* which occur after mast seeding of native beech and rimu (Kemp 2013). Monitoring by the Department of Conservation between 2009-2014 found that only 2% of kea nests in areas without pest control were successful, in contrast with a 27% success rate in areas treated with aerial 1080 in 2015 (Department of Conservation 2016). It is estimated that around 60% of kea nests are normally attacked by predators, especially stoats, which may also kill adults; this can rise to as many as 99% of nests being attacked in a stoat “plague” (Department of Conservation 2016). Birds are sometimes killed through accidents with human objects (e.g. motor vehicles, snow groomers, rubbish bins, electricity sub-stations) (Orr-Walker *et al.* 2015). Ingestion of lead from building components and the 1080 toxin used in invasive control have potentially widespread impacts on the population (T. Orr-Walker *in litt.* 2008; Reid *et al.* 2012; Orr-Walker 2013a; van Klink & Crowell 2015). Climate change may pose a threat through possible future

influences on its high altitude habitat (Mosen 2009).

## **Conservation Actions (see Appendix for additional information)**

### **Conservation Actions Underway**

CITES Appendix II. The species has been fully protected by national law since 1986 (Orr-Walker *et al.* 2015). Research is being conducted on its ecology and population dynamics. Advocacy is aimed at informing alpine users of ways to minimise adverse impacts and to change the negative image of the species often held by high-country farmers and ski-field operators (A. Grant *in litt.* 1999). The Kea Conservation Trust operates a conflict resolution programme (Orr-Walker *in litt.* 2016). Predator control has been carried out and there are plans for it to continue as part of New Zealand's "Battle for our Birds" (Department of Conservation 2016). Efforts to use a bird repellent to deter kea from 1080 toxin have not so far proved effective (van Klink and Crowell 2015). A project to involve communities in kea conservation is underway (Orr-Walker *in litt.* 2016).

### **Conservation Actions Proposed**

Census the population at frequent intervals. Research population dynamics and genetics (Orr-Walker *et al.* 2015). Continue advocacy campaigns. Continue to control introduced mammals. Identify sources of lead posing a risk to kea, then either remove, replace with lead free alternatives or cover to prevent kea access (Orr-Walker 2013a). Ensure that kea are excluded from predator-control toxins and devices, and limit the use of risky devices (Orr-Walker 2013b). Investigate the feasibility of a captive or island insurance population (Orr-Walker *et al.* 2015).

## **Credits**

**Assessor(s):** BirdLife International

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## External Resources

For [Images and External Links to Additional Information](#), please see the Red List website.

# Appendix

## Habitats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

| Habitat                                                                  | Season       | Suitability | Major Importance? |
|--------------------------------------------------------------------------|--------------|-------------|-------------------|
| 1. Forest -> 1.4. Forest - Temperate                                     | Resident     | Suitable    | Yes               |
| 1. Forest -> 1.6. Forest - Subtropical/Tropical Moist Lowland            | Breeding     | Suitable    | Yes               |
| 3. Shrubland -> 3.7. Shrubland - Subtropical/Tropical High Altitude      | Breeding     | Suitable    | Yes               |
| 4. Grassland -> 4.4. Grassland - Temperate                               | Resident     | Suitable    | Yes               |
| 4. Grassland -> 4.7. Grassland - Subtropical/Tropical High Altitude      | Resident     | Suitable    | Yes               |
| 14. Artificial/Terrestrial -> 14.2. Artificial/Terrestrial - Pastureland | Non-breeding | Marginal    | -                 |
| 14. Artificial/Terrestrial -> 14.5. Artificial/Terrestrial - Urban Areas | Non-breeding | Suitable    | No                |

## Threats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

| Threat                                                                                                                                | Timing    | Scope                                               | Severity                   | Impact Score     |
|---------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------|----------------------------|------------------|
| 11. Climate change & severe weather -> 11.1. Habitat shifting & alteration                                                            | Future    | Unknown                                             | Unknown                    | Unknown          |
|                                                                                                                                       | Stresses: | 1. Ecosystem stresses -> 1.2. Ecosystem degradation |                            |                  |
| 12. Other options -> 12.1. Other threat                                                                                               | Ongoing   | Majority (50-90%)                                   | Slow, significant declines | Medium impact: 6 |
|                                                                                                                                       | Stresses: | 2. Species Stresses -> 2.1. Species mortality       |                            |                  |
| 2. Agriculture & aquaculture -> 2.3. Livestock farming & ranching -> 2.3.2. Small-holder grazing, ranching or farming                 | Ongoing   | Minority (50%)                                      | Unknown                    | Unknown          |
|                                                                                                                                       | Stresses: | 1. Ecosystem stresses -> 1.1. Ecosystem conversion  |                            |                  |
| 4. Transportation & service corridors -> 4.1. Roads & railroads                                                                       | Ongoing   | Minority (50%)                                      | Unknown                    | Unknown          |
|                                                                                                                                       | Stresses: | 2. Species Stresses -> 2.1. Species mortality       |                            |                  |
| 5. Biological resource use -> 5.1. Hunting & trapping terrestrial animals -> 5.1.2. Unintentional effects (species is not the target) | Ongoing   | Minority (50%)                                      | Unknown                    | Unknown          |
|                                                                                                                                       | Stresses: | 2. Species Stresses -> 2.1. Species mortality       |                            |                  |
| 5. Biological resource use -> 5.1. Hunting & trapping terrestrial animals -> 5.1.3. Persecution/control                               | Ongoing   | Minority (50%)                                      | Negligible declines        | Low impact: 4    |
|                                                                                                                                       | Stresses: | 2. Species Stresses -> 2.1. Species mortality       |                            |                  |
| 6. Human intrusions & disturbance -> 6.1. Recreational activities                                                                     | Ongoing   | Unknown                                             | Unknown                    | Unknown          |

|                                                                                                                                                                         |           |                                                                                                                                                                                                                                  |                            |                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------|
|                                                                                                                                                                         | Stresses: | 2. Species Stresses -> 2.1. Species mortality                                                                                                                                                                                    |                            |                  |
| 8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species ( <i>Felis catus</i> )           | Ongoing   | Majority (50-90%)                                                                                                                                                                                                                | Slow, significant declines | Medium impact: 6 |
|                                                                                                                                                                         | Stresses: | 2. Species Stresses -> 2.1. Species mortality<br>2. Species Stresses -> 2.3. Indirect species effects -> 2.3.7. Reduced reproductive success                                                                                     |                            |                  |
| 8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species ( <i>Trichosurus vulpecula</i> ) | Ongoing   | Majority (50-90%)                                                                                                                                                                                                                | Slow, significant declines | Medium impact: 6 |
|                                                                                                                                                                         | Stresses: | 1. Ecosystem stresses -> 1.2. Ecosystem degradation<br>2. Species Stresses -> 2.3. Indirect species effects -> 2.3.2. Competition<br>2. Species Stresses -> 2.3. Indirect species effects -> 2.3.7. Reduced reproductive success |                            |                  |
| 8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species ( <i>Mustela erminea</i> )       | Ongoing   | Majority (50-90%)                                                                                                                                                                                                                | Rapid declines             | Medium impact: 7 |
|                                                                                                                                                                         | Stresses: | 2. Species Stresses -> 2.1. Species mortality<br>2. Species Stresses -> 2.3. Indirect species effects -> 2.3.7. Reduced reproductive success                                                                                     |                            |                  |

## Conservation Actions in Place

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

|                                                           |
|-----------------------------------------------------------|
| <b>Conservation Actions in Place</b>                      |
| In-Place Research, Monitoring and Planning                |
| Action Recovery plan: Yes                                 |
| Systematic monitoring scheme: No                          |
| In-Place Land/Water Protection and Management             |
| Conservation sites identified: Yes, over part of range    |
| Occur in at least one PA: Yes                             |
| Invasive species control or prevention: Yes               |
| In-Place Species Management                               |
| Harvest management plan: No                               |
| Successfully reintroduced or introduced benignly: No      |
| Subject to ex-situ conservation: No                       |
| In-Place Education                                        |
| Subject to recent education and awareness programmes: Yes |
| Included in international legislation: No                 |

|                                                             |
|-------------------------------------------------------------|
| <b>Conservation Actions in Place</b>                        |
| Subject to any international management/trade controls: Yes |

## Conservation Actions Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

|                                                                                                      |
|------------------------------------------------------------------------------------------------------|
| <b>Conservation Actions Needed</b>                                                                   |
| 2. Land/water management -> 2.2. Invasive/problematic species control                                |
| 3. Species management -> 3.4. Ex-situ conservation -> 3.4.1. Captive breeding/artificial propagation |
| 4. Education & awareness -> 4.3. Awareness & communications                                          |

## Research Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

|                                                            |
|------------------------------------------------------------|
| <b>Research Needed</b>                                     |
| 1. Research -> 1.2. Population size, distribution & trends |
| 3. Monitoring -> 3.1. Population trends                    |

## Additional Data Fields

|                                                                 |
|-----------------------------------------------------------------|
| <b>Distribution</b>                                             |
| Continuing decline in area of occupancy (AOO): Unknown          |
| Extreme fluctuations in area of occupancy (AOO): No             |
| Estimated extent of occurrence (EOO) (km <sup>2</sup> ): 119000 |
| Continuing decline in extent of occurrence (EOO): Unknown       |
| Extreme fluctuations in extent of occurrence (EOO): No          |
| Number of Locations: 11-100                                     |
| Continuing decline in number of locations: Unknown              |
| Extreme fluctuations in the number of locations: No             |
| Upper elevation limit (m): 2400                                 |
| <b>Population</b>                                               |
| Number of mature individuals: 4000                              |
| Continuing decline of mature individuals: Yes                   |
| Extreme fluctuations: No                                        |
| Population severely fragmented: No                              |

|                                                                       |
|-----------------------------------------------------------------------|
| <b>Population</b>                                                     |
| No. of subpopulations: 1                                              |
| Continuing decline in subpopulations: Unknown                         |
| Extreme fluctuations in subpopulations: No                            |
| All individuals in one subpopulation: Yes                             |
| No. of individuals in largest subpopulation: 100                      |
| <b>Habitats and Ecology</b>                                           |
| Continuing decline in area, extent and/or quality of habitat: Unknown |
| Generation Length (years): 12                                         |
| Movement patterns: Not a Migrant                                      |

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